6. Python Program to Solve the Vacuum Cleaner Problem

Aim: To implement the Vacuum Cleaner Problem using Python and demonstrate how an agent cleans dirty rooms.

Procedure:

1. Define two rooms, A and B, with their initial states.
2. Represent each room as either Clean or Dirty.
3. Set the vacuum cleaner's initial position.
4. Check whether the current room is dirty.
5. Clean the dirty room and move to the next room.
6. Display the final state of both rooms.

```
rooms = {'A': 'Dirty', 'B': 'Dirty'}
position = 'A'

while True:
    print("Vacuum is in room", position)

    if rooms[position] == 'Dirty':
        rooms[position] = 'Clean'
        print("Room", position, "cleaned")

    if position == 'A':
        position = 'B'
    else:
        break

print("Final Room States:", rooms)
```

Output:

```
Python 3.14.7 (tags/v3.14.7:823f032, Aug  5 2026,
AMD64)] on win32
Enter "help" below or click "Help" above for more
>>
========== RESTART: C:/Users/acer/Downloads/C
Vacuum is in room A
Room A cleaned
Vacuum is in room B
Room B cleaned
Final Room States: {'A': 'Clean', 'B': 'Clean'}
>>
```

Result: Thus, the Vacuum Cleaner Problem was implemented successfully using Python, and both rooms were cleaned.